Reproduction of Table I of arXiv:1708.02243

Knapen, Lou, Papucci, Setford, Phys. Rev. D 96, 115015 (2017) · 18 October 2026 · production samples 2000 events each · EW leading-order samples: 3000 events (no shower), 2792 (with shower)

Questions to answer

- **Q1. Why is the converted-photon upper limit the same for both cross sections?** → Section 7.
- **Q2. Figure 2: why does our |L| go to zero at every second meeting while the paper's does not?** → Section 3 (with Appendix A of arXiv:2002.07503).
- **Q3. Which smearing was applied, and how?** → Section 1.
- **Q4. Which uncertainties are quoted?** → Section 1 (formula) and every table (k/n = ε ± σ).

1. How the hits are smeared, and the uncertainty used

Detector part	Formula applied to the true crossing point	Value
Pixel layers (4)	hit moved to the centre of its pixel: $x_{meas} = p \cdot \mathrm{round}(x / p)$	p = 50 μm (rφ); 250 μm (IBL z), 400 μm (other z)
SCT layers (4), rφ	$(r\varphi)_{meas} = r\varphi + \sigma_{r\varphi} \cdot g$, g = Gaussian random number (mean 0, width 1)	$\sigma_{r\varphi}$ = 34 μm (2 × 17 μm)
SCT layers (4), z	$z_{meas} = z + \sigma_z \cdot g$	σ_z = 0 (“no smearing”) or 1.16 mm (2 × 580 μm)
Beam spot	all hits of an event shifted by the same z_0, z_0 = 45 mm · g	45 mm
Neighbouring pixels	hits in adjacent pixels of one layer merged into one hit	—

Pixel rounding gives an RMS error p/√12 (14 μm for 50 μm, 115 μm for 400 μm). Everything is from the paper (Sec. III A) and Ref. [27], except σ_z = 0 (our choice: it is the only value that reproduces the paper's numbers) and one common z_0 per event (our reading).

Uncertainty of an efficiency ε = k / n: $\sigma = \sqrt{\varepsilon\,(1-\varepsilon)\,/\,n}$ (statistical only)

Decision: **consistent** with the paper if $|\varepsilon - \varepsilon_{paper}| \le 2\,\sigma_0$, $\sigma_0 = \sqrt{\varepsilon_{paper}\,(1-\varepsilon_{paper})\,/\,n}$

2. Table I

2.1 Efficiencies at every Λ

Colored quirk, m = 1800 GeV (2000 events generated, 873 after matching)

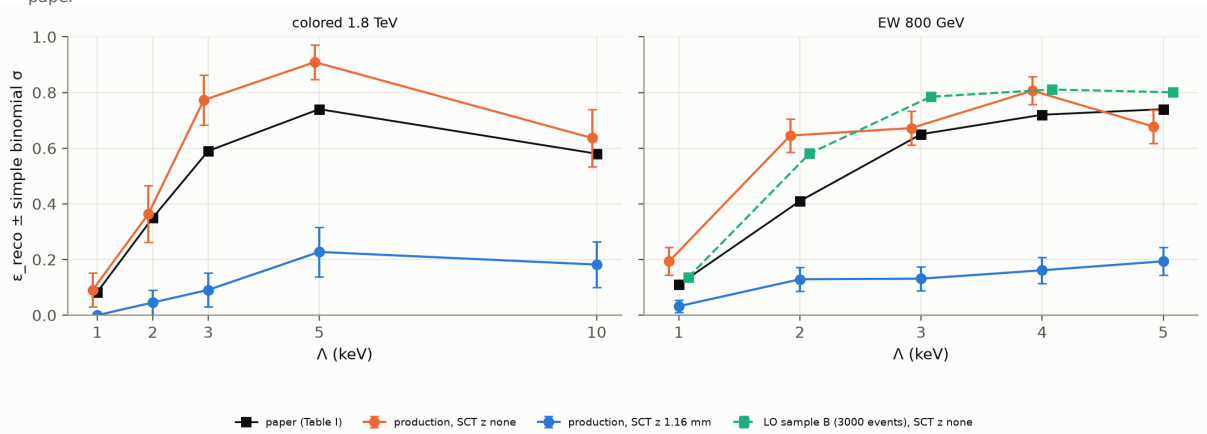
Λ (keV)	ε_{trig} = N(p_T(QQ) > 200 GeV) / N(matched)	ε_{fid} = N(fiducial) / N(trig., both charged)	ε_{reco} = N(signal region) / N(fiducial), no smearing	paper ε_{reco}	decision	ε_{reco}, SCT z 1.16 mm	decision
1	207/873 = 0.24 ± 0.01	22/68 = 0.32 ± 0.06	2/22 = 0.09 ± 0.06	0.083	consistent |Δ| 0.008 ≤ 2σ₀ 0.118	0/22 = 0.00 (95% UL 0.13)	consistent |Δ| 0.083 ≤ 2σ₀ 0.118
2	207/873 = 0.24 ± 0.01	22/68 = 0.32 ± 0.06	8/22 = 0.36 ± 0.10	0.35	consistent |Δ| 0.014 ≤ 2σ₀ 0.203	1/22 = 0.05 ± 0.04	**inconsistent** |Δ| 0.305 > 2σ₀ 0.203
3	207/873 = 0.24 ± 0.01	22/68 = 0.32 ± 0.06	17/22 = 0.77 ± 0.09	0.59	consistent |Δ| 0.183 ≤ 2σ₀ 0.210	2/22 = 0.09 ± 0.06	**inconsistent** |Δ| 0.499 > 2σ₀ 0.210
5	207/873 = 0.24 ± 0.01	22/68 = 0.32 ± 0.06	20/22 = 0.91 ± 0.06	0.74	consistent |Δ| 0.169 ≤ 2σ₀ 0.187	5/22 = 0.23 ± 0.09	**inconsistent** |Δ| 0.513 > 2σ₀ 0.187
10	207/873 = 0.24 ± 0.01	22/68 = 0.32 ± 0.06	14/22 = 0.64 ± 0.10	0.58	consistent |Δ| 0.056 ≤ 2σ₀ 0.210	4/22 = 0.18 ± 0.08	**inconsistent** |Δ| 0.398 > 2σ₀ 0.210
paper	0.24	0.28	see column				
dec.	consistent |Δ| 0.003 ≤ 2σ₀ 0.029	consistent |Δ| 0.044 ≤ 2σ₀ 0.109					

EW quirk, m = 800 GeV (2000 events generated, 1045 after matching)

Λ (keV)	ε_{trig} = N(p_T(QQ) > 200 GeV) / N(matched)	ε_{fid} = N(fiducial) / N(trig.)	ε_{reco} = N(signal region) / N(fiducial), no smearing	paper ε_{reco}	decision	ε_{reco}, SCT z 1.16 mm	decision
1	186/1045 = 0.18 ± 0.01	62/186 = 0.33 ± 0.03	12/62 = 0.19 ± 0.05	0.11	**inconsistent** |Δ| 0.084 > 2σ₀ 0.079	2/62 = 0.03 ± 0.02	consistent |Δ| 0.078 ≤ 2σ₀ 0.079
2	186/1045 = 0.18 ± 0.01	62/186 = 0.33 ± 0.03	40/62 = 0.65 ± 0.06	0.41	**inconsistent** |Δ| 0.235 > 2σ₀ 0.125	8/62 = 0.13 ± 0.04	**inconsistent** |Δ| 0.281 > 2σ₀ 0.125

Λ (keV)	ε_{trig} = N(p_T(QQ) > 200 GeV) / N(matched)	ε_{fid} = N(fiducial) / N(trig.)	ε_{reco} = N(signal region) / N(fiducial), no smearing	paper ε_{reco}	decision	ε_{reco}, SCT z 1.16 mm	decision
3	186/1045 = 0.18 ± 0.01	61/186 = 0.33 ± 0.03	41/61 = 0.67 ± 0.06	0.65	consistent |Δ| 0.022 ≤ 2σ₀ 0.122	8/61 = 0.13 ± 0.04	**inconsistent** |Δ| 0.519 > 2σ₀ 0.122
4	186/1045 = 0.18 ± 0.01	62/186 = 0.33 ± 0.03	50/62 = 0.81 ± 0.05	0.72	consistent |Δ| 0.086 ≤ 2σ₀ 0.114	10/62 = 0.16 ± 0.05	**inconsistent** |Δ| 0.559 > 2σ₀ 0.114
5	186/1045 = 0.18 ± 0.01	62/186 = 0.33 ± 0.03	42/62 = 0.68 ± 0.06	0.74	consistent |Δ| 0.063 ≤ 2σ₀ 0.111	12/62 = 0.19 ± 0.05	**inconsistent** |Δ| 0.546 > 2σ₀ 0.111
paper	0.10	0.28	see column				
dec.	**inconsistent** |Δ| 0.078 > 2σ₀ 0.019	consistent |Δ| 0.053 ≤ 2σ₀ 0.066					

ε_{fid} is recomputed at every Λ. EW at 3 keV: 61 instead of 62 because one event (no. 127) could not be integrated at 3 keV (the solver did not reach 300 steps per oscillation within its step limit); like every such failure it is counted as not fiducial. It is fiducial at 1, 2, 4 and 5 keV. For k = 0 the binomial σ is zero, so the one-sided 95% upper limit $1 - 0.05^{1/n}$ is given. Decision cells give $|\Delta| = |\varepsilon - \varepsilon_{paper}|$ and $2\sigma_0$ with the n of that row.



What we learn. Colored: no smearing consistent at every Λ; 1.16 mm inconsistent at Λ = 2, 3, 5, 10 keV. EW: no smearing inconsistent at Λ = 1, 2 keV (higher than the paper), consistent elsewhere; 1.16 mm inconsistent at Λ = 2, 3, 4, 5 keV. The SCT z smearing decides the result: without it our ε_{reco} is close to the paper, with the paper's 1.16 mm it is far below.

2.2 Event counts at every Λ (cut-flow, first crossing)

Colored 1800 GeV: matched 873 → triggered 207 → both charged 68 (same for every Λ).

Λ (keV) →	**1**		**2**		**3**		**5**		**10**	
step ↓ / SCT z →	**none**	**1.16**	**none**	**1.16**	**none**	**1.16**	**none**	**1.16**	**none**	**1.16**
fiducial (both quirks cross all 8 layers, t < 25 ns)	22	22	22	22	22	22	22	22	22	22
plane found (Δw < 1 cm)	4	1	18	9	21	15	22	19	22	22
+ hit multiplicity (2 hits in 7 layers, ≥ 1 in 8th)	2	0	8	4	17	5	20	9	14	8
+ Δs < 0.01 cm = signal region (pass)	**2**	**0**	**8**	**1**	**17**	**2**	**20**	**5**	**14**	**4**
fail (fiducial − pass)	20	22	14	21	5	20	2	17	8	18
ε_{reco} = pass / fiducial	0.09	0.00	0.36	0.05	0.77	0.09	0.91	0.23	0.64	0.18
$\varepsilon = \varepsilon_{trig} \cdot \varepsilon_{fid} \cdot \varepsilon_{reco}$ (Eq. 12), ×10⁻²	**0.70**	**0.00**	**2.79**	**0.35**	**5.93**	**0.70**	**6.97**	**1.74**	**4.88**	**1.39**
paper ε ×10⁻² (= paper $\varepsilon_{trig}\cdot\varepsilon_{fid}\cdot\varepsilon_{reco}$)	0.56		2.35		3.96		4.97		3.90	

EW 800 GeV: matched 1045 → triggered 186 (same for every Λ).

Λ (keV) →	**1**		**2**		**3**		**4**		**5**	
step ↓ / SCT z →	**none**	**1.16**	**none**	**1.16**	**none**	**1.16**	**none**	**1.16**	**none**	**1.16**
fiducial (both quirks cross all 8 layers, t < 25 ns)	62	62	62	62	61	61	62	62	62	62
plane found (Δw < 1 cm)	21	14	54	36	61	40	62	45	62	49
+ hit multiplicity (2 hits in 7 layers, ≥ 1 in 8th)	12	7	40	21	41	21	50	24	42	24

Λ (keV) →	**1**		**2**		**3**		**4**		**5**	
+ Δs < 0.01 cm = signal region (pass)	**12**	**2**	**40**	**8**	**41**	**8**	**50**	**10**	**42**	**12**
fail (fiducial − pass)	50	60	22	54	20	53	12	52	20	50
ε_{reco} = pass / fiducial	0.19	0.03	0.65	0.13	0.67	0.13	0.81	0.16	0.68	0.19
$\varepsilon = \varepsilon_{trig} \cdot \varepsilon_{fid} \cdot \varepsilon_{reco}$ (Eq. 12), $\times 10^{-2}$	**1.15**	**0.19**	**3.83**	**0.77**	**3.92**	**0.77**	**4.78**	**0.96**	**4.02**	**1.15**
paper $\varepsilon \times 10^{-2}$ (= paper $\varepsilon_{trig} \cdot \varepsilon_{fid} \cdot \varepsilon_{reco}$)	0.31		1.15		1.82		2.02		2.07	

"none" = no SCT z smearing, "1.16" = SCT z 1.16 mm. ε_{fid} = fiducial / 68 (colored, both charged) or 186 (EW). The paper does not quote ε; its value here is the product of its Table I entries.

2.3 EW benchmark with leading-order samples

Reconstruction efficiency ε_{reco} = N(signal region) / N(fiducial), no SCT z smearing, first crossing

Λ (keV)	**Production: MadGraph + Pythia, matched (2000 generated)**	**LO, no shower (3000 events)**	**LO + Pythia shower (2792 events)**	**paper ε_{reco}**
1	12/62 = 0.19 ± 0.05 **inconsistent** $\lvert\Delta\rvert$ 0.084 > $2\sigma_0$ 0.079	121/885 = 0.14 ± 0.01 **inconsistent** $\lvert\Delta\rvert$ 0.027 > $2\sigma_0$ 0.021	127/950 = 0.13 ± 0.01 **inconsistent** $\lvert\Delta\rvert$ 0.024 > $2\sigma_0$ 0.020	0.11
2	40/62 = 0.65 ± 0.06 **inconsistent** $\lvert\Delta\rvert$ 0.235 > $2\sigma_0$ 0.125	520/895 = 0.58 ± 0.02 **inconsistent** $\lvert\Delta\rvert$ 0.171 > $2\sigma_0$ 0.033	548/955 = 0.57 ± 0.02 **inconsistent** $\lvert\Delta\rvert$ 0.164 > $2\sigma_0$ 0.032	0.41
3	41/61 = 0.67 ± 0.06 consistent $\lvert\Delta\rvert$ 0.022 ≤ $2\sigma_0$ 0.122	707/901 = 0.78 ± 0.01 **inconsistent** $\lvert\Delta\rvert$ 0.135 > $2\sigma_0$ 0.032	750/953 = 0.79 ± 0.01 **inconsistent** $\lvert\Delta\rvert$ 0.137 > $2\sigma_0$ 0.031	0.65
4	50/62 = 0.81 ± 0.05 consistent $\lvert\Delta\rvert$ 0.086 ≤ $2\sigma_0$ 0.114	727/897 = 0.81 ± 0.01 **inconsistent** $\lvert\Delta\rvert$ 0.090 > $2\sigma_0$ 0.030	779/959 = 0.81 ± 0.01 **inconsistent** $\lvert\Delta\rvert$ 0.092 > $2\sigma_0$ 0.029	0.72
5	42/62 = 0.68 ± 0.06 consistent $\lvert\Delta\rvert$ 0.063 ≤ $2\sigma_0$ 0.111	719/898 = 0.80 ± 0.01 **inconsistent** $\lvert\Delta\rvert$ 0.061 > $2\sigma_0$ 0.029	746/961 = 0.78 ± 0.01 **inconsistent** $\lvert\Delta\rvert$ 0.036 > $2\sigma_0$ 0.028	0.74

Trigger and fiducial efficiency

	Production	**LO, no shower**	**LO + Pythia shower**	**paper**
ε_{trig}	186/1045 = 0.18 ± 0.01 **inconsistent** $\lvert\Delta\rvert$ 0.078 > $2\sigma_0$ 0.019	σ_B/σ_A = 0.096 ± 0.001 consistent $\lvert\Delta\rvert$ 0.004 ≤ max(2σ, 0.005)	0.096 × 2792/3000 = 0.089 ± 0.001 **inconsistent** $\lvert\Delta\rvert$ 0.011 > max(2σ, 0.005)	0.10
ε_{fid} (4 keV)	62/186 = 0.33 ± 0.03 consistent $\lvert\Delta\rvert$ 0.053 ≤ $2\sigma_0$ 0.066	897/3000 = 0.30 ± 0.01 **inconsistent** $\lvert\Delta\rvert$ 0.019 > $2\sigma_0$ 0.016	959/2792 = 0.34 ± 0.01 **inconsistent** $\lvert\Delta\rvert$ 0.063 > $2\sigma_0$ 0.017	0.28

For the cross-section ratios the statistical error (±0.001) is smaller than the precision of the paper's value (0.10, two decimals), so these two cells use $\lvert\Delta\rvert \le \max(2\sigma, 0.005)$; all other cells use the $2\sigma_0$ rule of Section 1.

How the two leading-order samples were made

- **Why.** Our trigger efficiency for the EW benchmark (0.18) is far above the paper's 0.10. arXiv:1911.02223 (Appendix A) obtains a trigger efficiency of about 0.1 as a leading-order cross-section ratio, σ(pp → QQ̄ + jet, p_T(jet) > 200 GeV) / σ(pp → QQ̄). The paper does not state how its value was obtained, so we computed the EW benchmark in this way.
- **Generation.** MadGraph5_aMC@NLO 3.7.3, quirk model, pp at 13 TeV, PDF nn23lo1, m_Q = 800 GeV. Sample A: pp → QQ̄, σ_A = 0.2307 fb. Sample B: pp → QQ̄ + jet with p_T(jet) > 200 GeV, σ_B = 0.02214 fb, 3000 unweighted events. $\varepsilon_{trig} = \sigma_B/\sigma_A$ = 0.096 ± 0.001; MadGraph 2.3.3 (the version in the paper's source) gives 0.096.
- **LO, no shower:** the quirk momenta are taken directly from the 3000 events of sample B. **LO + shower:** the same events are showered with Pythia 8.317 (no matching); 2792 of 3000 keep p_T(QQ̄) > 200 GeV after the shower, so $\varepsilon_{trig} = \sigma_B/\sigma_A \times 2792/3000$.
- **Everything else is as for the production sample:** detector model, Λ values, trajectory simulation, fiducial definition, plane search and signal region, first crossing, no SCT z smearing. ε_{fid} = N(fiducial) / N(events in the sample).
- **Numerical failures:** 122 of 15000 event–Λ trajectories (LO) and 111 of 13960 (LO + shower) could not be integrated; they are counted as not fiducial, as in the production sample.

Result. Without shower the leading-order ratio gives ε_{trig} = 0.096, consistent with the paper's 0.10; after the shower 2792/3000 events keep p_T(QQ̄) > 200 GeV and ε_{trig} = 0.089 (inconsistent). So the paper's 0.10 corresponds to the leading-order ratio, not to our matched sample (0.18). ε_{reco} is almost the same in all three samples (0.58 / 0.57 at 2 keV, 0.81 / 0.81 at 4 keV, LO / LO + shower) and is above the paper at Λ = 1, 2, 3, 4, 5 keV in the LO sample. With about 800–900 fiducial events the uncertainty is ±0.01–0.02, so this is not a statistical fluctuation, and it does not depend on how the events were generated. For the colored benchmark the same ratio gives 0.14; the matched production sample already agrees with the paper (0.24).

3. Figure 2 and Appendix A of arXiv:2002.07503

Equations used (the paper's model, solved numerically):

Eq. 3: $d(m\gamma\mathbf{v})/dt = -\Lambda^2 [\sqrt{(1-v_\perp^2)}\,\hat{s} + (v_\parallel/\sqrt{(1-v_\perp^2)})\,\mathbf{v}_\perp] + q\,\mathbf{v}\times\mathbf{B}$ (string + magnetic force)

$L = (\mathbf{x}_1^* - \mathbf{x}_2^*) \times (\mathbf{p}_1^* - \mathbf{p}_2^*) / 2$ in the pair's rest (CM) frame at equal CM time; $d = |\mathbf{x}_1 - \mathbf{x}_2|$; r = distance of the pair's centre from the beam

Eq. 5: in the pair's rest frame the magnetic field acts as an electric field $\mathbf{E}^* = \gamma_{cm}\,\mathbf{v}_{cm} \times \mathbf{B}$; torque $\boldsymbol{\tau} = q\,\mathbf{d} \times \mathbf{E}^*$ (paper: $\tau \sim 2\mathbf{d} \times e\mathbf{E}_{cm}$, d = one quirk's distance from the CM)

What Appendix A of arXiv:2002.07503 shows (Li, Li, Pei, Zhang, "The quirk trajectory", same equation, uniform magnetic field):

- Eqs. (II.11)–(II.14): in the pair's rest frame the two quirks move back and forth on one line and meet at every integer g (g = $\Lambda^2 t/(2m\rho)$ is the time in units of half an oscillation).
- Text after Eq. (A.4): "the average torque on the system of two quirks is zero when g is increased from 2n to 2n + 2 such that the total angular momentum of the quirk pair is also invariant at g = 2n".
- Eq. (A.5): $z_1(2n) = -z_2(2n) = 0$, i.e. at every second meeting the quirks meet head-on, as at production.

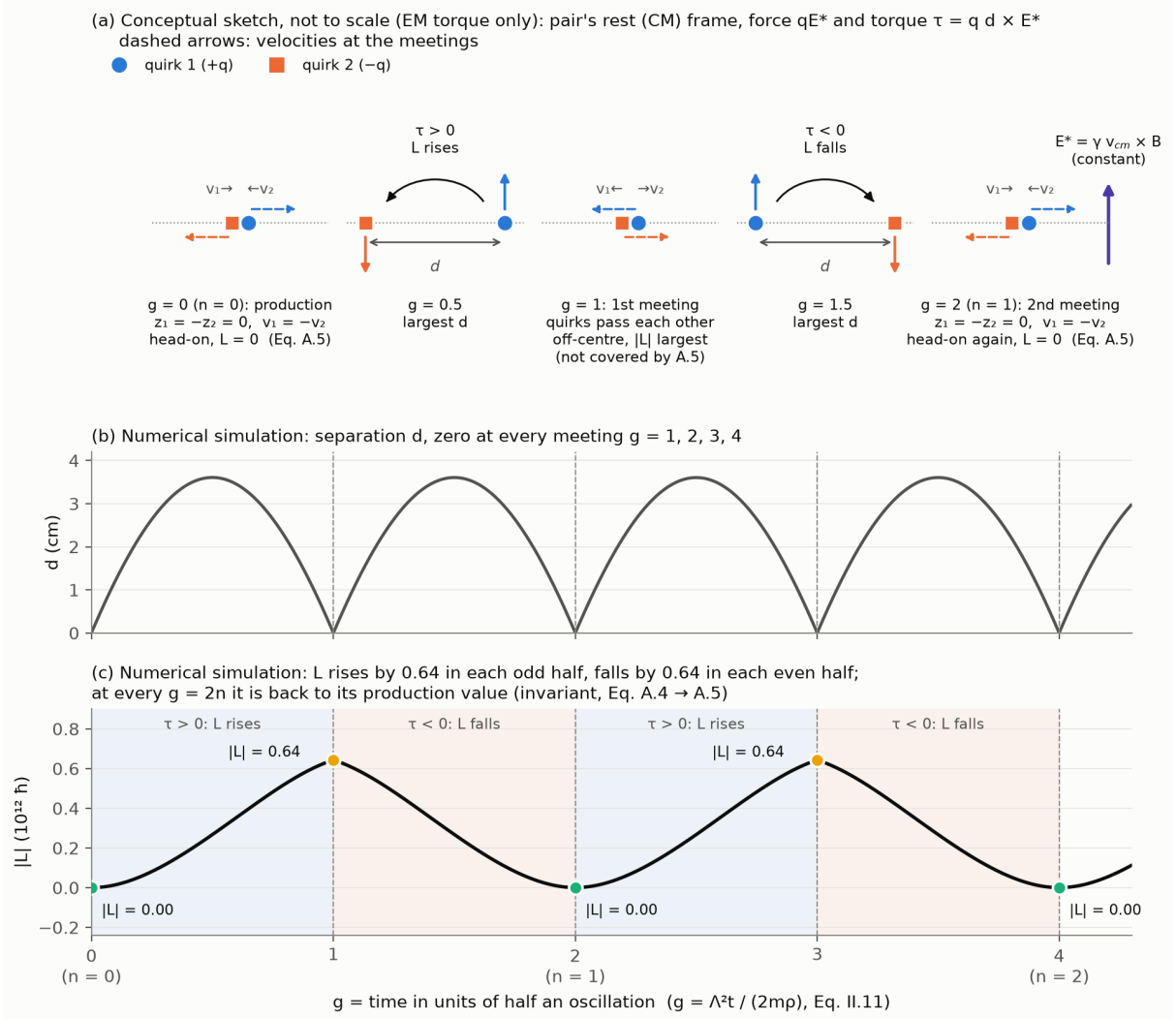


Figure: how L builds up and returns. (a) is a **conceptual sketch** (not to scale, EM torque only; not simulation output) of the mechanism and of the relation of Eq. A.5 at the head-on meetings g = 0 and 2: $z_1 = -z_2 = 0$ with opposite velocities $v_1 = -v_2$. Eq. A.5 does not cover the odd meetings, where the quirks pass off-centre with the largest |L|. (b) and (c) are the **numerical simulation** of Eq. 3 with the Lorentz force for the event with the paper's kinematics: |L| = 0.64 ×10¹² ħ at g = 1, 3 and consistent with zero at g = 2, 4. The agreement with Eq. A.4–A.5 is the numerical result; (a) only illustrates it.

Why. Between two meetings the separation **d** points one way, in the next half-oscillation the other way. The field **E*** is constant, so the torque $q\mathbf{d}\times\mathbf{E}^*$ has the same size and the opposite sign in the two halves: L goes up, then back to zero. Our simulation shows exactly this.

Change of L in each half-oscillation (event with the paper's kinematics, units 10¹² ħ)	**1st**	**2nd**	**3rd**	**4th**
total change of L in our simulation	+0.64	-0.64	+0.64	-0.64
from the magnetic force (torque $q\mathbf{d}\times\mathbf{E}^*$)	+0.56	-0.56	+0.56	-0.56
from the string's own term in Eq. 3	+0.09	-0.09	+0.09	-0.09
paper's Fig. 2 (read from its curves)	**+0.65**	**-0.84**	**+0.71**	**-0.80**

In our simulation each half-oscillation adds or removes the same amount (±0.64), so L returns to zero after every two halves. In the paper the halves are unequal (+0.65, −0.84, +0.71, −0.80), so L never returns to zero at a meeting.

m = 1800 GeV, Λ = 2 keV, B = 2 T along the beam, production at the origin, RK4 with 3000 steps per oscillation.

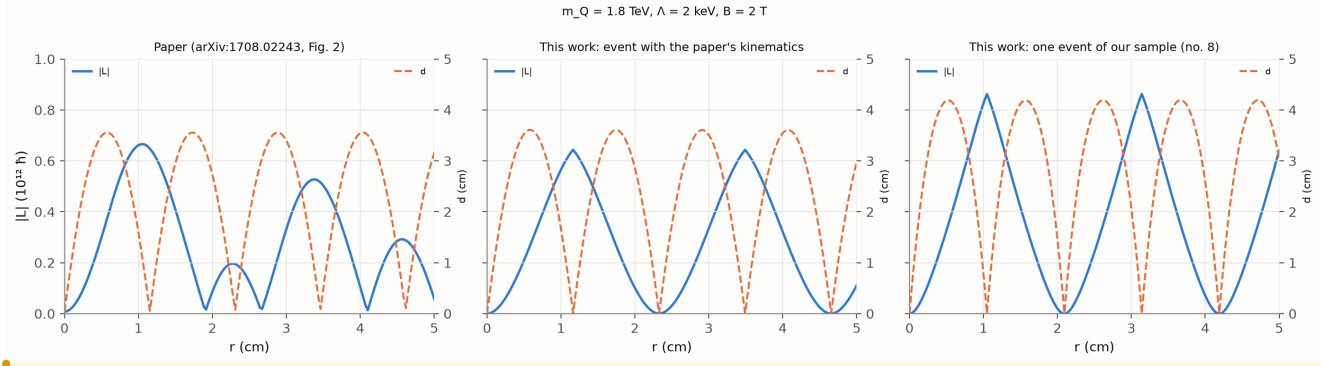
Initial values of our two simulated events	**Event with the paper's kinematics**	**One event of our sample (no. 8)**
charges	+2/3, −2/3	+1, -1
quirk 1 lab momentum (p_x, p_y, p_z)	212.2 , 0.0 , 1204.1 GeV	138.4 , -1123.0 , -424.7 GeV
quirk 2 lab momentum	212.2 , 0.0 , -1204.1 GeV	51.5 , 1430.4 , 144.6 GeV
pair speed β_{cm} (transverse)	0.0975	0.0808
internal speed v* (CM frame)	0.556	0.5874
p* = m γ* v*	1204.1 GeV	1307.0 GeV
predicted d max = 2(γ* − 1) m/Λ²	3.607 cm	4.185 cm
where the values come from	v* from the paper's d max (Eq. 4), β_{cm} from the spacing of its meetings; oscillation along the beam and charges ±2/3 chosen by us (the paper does not give them)	MadGraph 3.7.3 + Pythia 8.317 colored sample; chosen because its d max and spacing are nearest the paper's; charges of the final-state R-hadrons

Meeting by meeting (paper: from its vector figure; its signed L assumes a sign change at each zero of |L|)

meeting		**r (cm)**	**|L| (10^{12} ħ)**	**signed L**	**change of L in this half**	**|L| = 0 at r (cm)**	**b = |L| / p* (cm)**
1	**paper**	1.16	0.65	+0.65	**+0.64**	—	1.1×10^{-5}
	ours, paper's kinematics	1.16	0.64	+0.64	**+0.64**	—	1.1×10^{-5}
	ours, event no. 8	1.05	0.86	+0.86	**+0.86**	—	1.3×10^{-5}
2	**paper**	2.31	0.20	-0.20	**-0.84**	between meetings: 1.91	3.2×10^{-6}
	ours, paper's kinematics	2.33	0.00	+0.00	**-0.64**	at the meeting	6.9×10^{-9}†
	ours, event no. 8	2.10	0.00	+0.00	**-0.86**	at the meeting	2.1×10^{-9}†
3	**paper**	3.46	0.52	+0.52	**+0.71**	between meetings: 2.67	8.4×10^{-6}
	ours, paper's kinematics	3.49	0.64	+0.64	**+0.64**	—	1.1×10^{-5}
	ours, event no. 8	3.14	0.86	+0.86	**+0.86**	—	1.3×10^{-5}
4	**paper**	4.62	0.29	-0.29	**-0.80**	between meetings: 4.10	4.7×10^{-6}
	ours, paper's kinematics	4.65	0.00	+0.00	**-0.64**	at the meeting	2.2×10^{-9}†
	ours, event no. 8	4.19	0.00	+0.00	**-0.86**	at the meeting	7.4×10^{-9}†

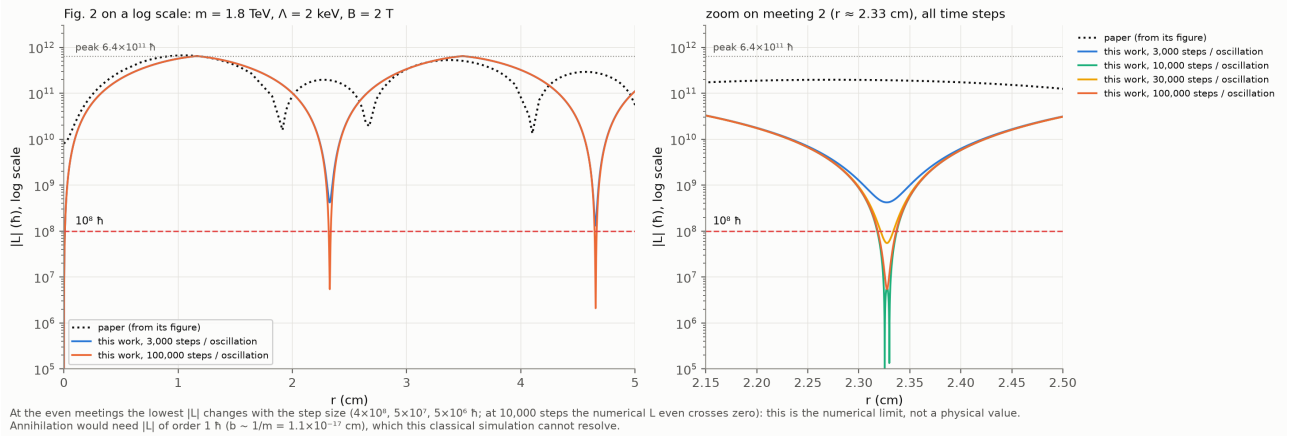
d max is the same in every half-oscillation for the paper (3.556 cm) and for us (3.61 cm, no energy loss). b uses p* = 1204 GeV for the paper (it gives no p*); $1/m_Q = 1.1\times10^{-17}$ cm. † At the even meetings our left-over |L| (~10^8 ħ) is the limit of the simulation and changes with the time step; it is not converged.

What the table shows. The first meeting agrees (0.65 vs 0.64). In our simulation every half-oscillation changes L by the same amount (±0.64; ±0.86 for event 8), so |L| is back to zero at every second meeting. In the paper the changes are unequal (+0.64, −0.84, +0.71, −0.80): L changes sign between the meetings, its large peaks fall (0.67 → 0.53) and its small ones rise (0.20 → 0.29). With the same d in every half and a uniform field, the paper's Eq. 3 and Lorentz force give equal changes; the paper does not state what else entered its calculation.



Conclusion. Our Fig. 2 is what the paper's own equation gives; the same result is found analytically in arXiv:2002.07503 (Appendix A) and with an independent code. The paper's later |L| values cannot be obtained from its stated equation; the paper does not give the event's charges and momenta or how L was computed.

3.1 Is L exactly zero? Fig. 2 on a log scale



At the even meetings the lowest |L| changes with the step size (4×10⁸, 5×10⁷, 5×10⁶ ħ; at 10,000 steps the numerical L even crosses zero): this is the numerical limit, not a physical value. Annihilation would need |L| of order 1 ħ (b ~ 1/m = 1.1×10⁻¹⁷ cm), which this classical simulation cannot resolve.

RK4 steps per oscillation	|L| at meeting 2 (ħ)	|L| at meeting 4 (ħ)	|L| at meetings 1 and 3 (ħ)
3,000	4.2×10^{8}	1.3×10^{8}	6.43×10^{11}
10,000	5.9×10^{6}	2.4×10^{7}	6.43×10^{11}
30,000	5.5×10^{7}	5.8×10^{7}	6.43×10^{11}
100,000	5.5×10^{6}	2.1×10^{6}	6.43×10^{11}

What the log scale shows. At the meetings 1 and 3, |L| = 6.43×10^{11} ħ for every step size (converged, physical). At the meetings 2 and 4 the left-over |L| changes with the step size ($4\times10^{8} \to 5\times10^{7} \to 5\times10^{6}$ ħ; at 10,000 steps it even crosses zero): it is the numerical limit, not a physical value. So: on the scale of Fig. 2 (10^{12} ħ) L returns to zero, as Appendix A of arXiv:2002.07503 predicts exactly; at the ħ level needed for annihilation the classical simulation cannot decide. The paper's 2×10^{11} ħ at meeting 2 is at least 500× above any of our values.

4. Does L = 0 mean the quirks annihilate?

Paper (Sec. II, text of Fig. 2): “whenever |L| = 0, the separation between the quirks is large”, so they do not annihilate. **This work and arXiv:2002.07503 (App. A):** |L| returns to its production value at every second meeting. So the paper's reason does not hold. Two points show why this still does not mean annihilation:

4.1 Classical L ≈ 0 is not a collision

- Classically L = b × p, where b is the distance by which the two quirks miss each other (impact parameter): b = |L| / p.
- To annihilate, the quirks must overlap at the quantum level, b ~ $1/m_Q = 1.1\times10^{-17}$ cm for m_Q = 1800 GeV. At that distance L is only about one ħ (b > $1/m_Q$ needs |L| > 0.67 ħ).
- In our simulation (and in the independent code) the left-over |L| at the even meetings is 10^{6}–10^{8} ħ, so the quirks still miss each other by millions of times $1/m_Q$:

time steps per oscillation	|L| at meeting 2 (ħ)	|L| at meeting 4 (ħ)	b = |L| / p (cm)	b / ($1/m_Q$)
3,000	4.2×10^{8}	1.3×10^{8}	2×10^{-9} – 7×10^{-9}	2×10^{8} – 6×10^{8}
10,000	5.9×10^{6}	2.4×10^{7}	1×10^{-10} – 4×10^{-10}	9×10^{6} – 4×10^{7}
30,000	5.5×10^{7}	5.8×10^{7}	9×10^{-10} – 9×10^{-10}	8×10^{7} – 9×10^{7}
100,000	5.5×10^{6}	2.1×10^{6}	3×10^{-11} – 9×10^{-11}	3×10^{6} – 8×10^{6}

Event with the paper's kinematics, p* = 1204 GeV. A classical simulation cannot resolve L of a few ħ: the left-over L changes with the time step and does not converge, so we do not know where it would stop.

4.2 Even if L returns exactly to its production value, annihilation is not certain

- arXiv:2002.07503 (App. A, Eq. A.5): at every second meeting the quirks meet head-on, as at production, where the pair's angular momentum is zero (arXiv:2108.06748, Sec. 4: “the initial L'_{tot} at the IP is zero”). With this symmetry annihilation becomes **possible**.
- Possible is not the same as happening. The annihilation cross section of heavy quirks is small, $\sigma v \propto \alpha^2/m_Q^2$ (arXiv:2404.13814, Eq. 4.11). Worst case, if an annihilation chance ≈ $\alpha^2\gamma v$ is allowed at every second meeting (α = 0.02 / 0.05), the fraction of pairs surviving inside the tracker is 0.995 / 0.970 (colored, 2 keV), 0.971 / 0.835 (EW, 4 keV) and 0.887 / 0.494 (colored, 10 keV): small at the benchmarks, not small at large Λ.
- The symmetry is not perfect in a real detector. arXiv:2108.06748 (Sec. 4) shows that ionisation in material gives the pair angular momentum, and L changes sharply when the pair enters or leaves material, or when only one quirk is inside it. Tracker layers are material. Radiation and a non-uniform field are also absent from the paper's model and ours. (The turning angle η of arXiv:2002.07503, Eq. A.6, rotates the oscillation but keeps the head-on meetings of Eq. A.5, so η alone does not stop L from returning.)
- arXiv:2404.13814 models annihilation only after the pair has radiated its energy and dropped into the quirkonium ground state, not during the oscillations. It finds that radiative energy loss matters only for Λ ≳ 100 keV, far above the keV values here.

Conclusion. L returning to zero (this work, Appendix A) removes the paper's argument, but it does not show that the quirks annihilate. That needs a quantum estimate of the per-meeting probability, which neither the paper nor this study provides. Table I assumes no annihilation.

5. Quirks crossing the same layer more than once

Sample (fiducial events, all crossings within 25 ns)	events with > 2 hits on at least one layer	fraction ± σ
Colored 1800 GeV, 2 keV	13 / 22	0.59 ± 0.10
EW 800 GeV, 4 keV	31 / 62	0.50 ± 0.06

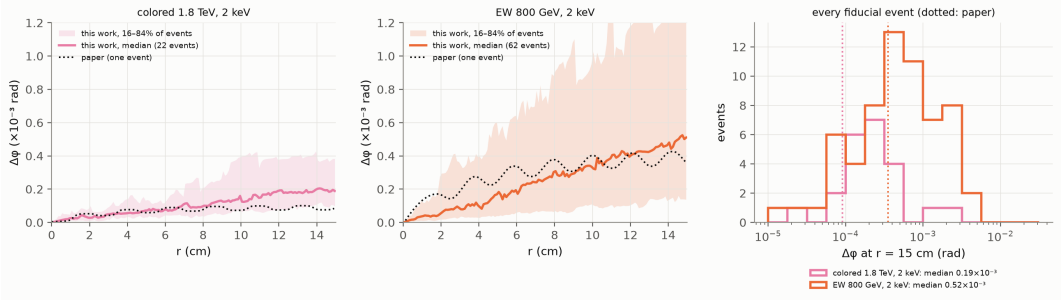
These are results of our simulation. The paper says each quirk “will most likely intersect each layer only once”. arXiv:2002.07503 (Sec. VI, Table I and Appendix D, Table II) also finds repeated crossings, more often at larger Λ and larger m/p_T, with 4–8 hits on some layers at Λ = 2 keV; its percentages (~10%) are for other masses and the CMS tracker and are not directly comparable. Table I above uses the first crossing only.

6. Figures 3, 4, 5

Figure 3: turning of the quirks' plane

Δφ = angle between the plane of the two velocity vectors at radius r and at production (paper's definition)

Fig. 3: Δφ at r = 15 cm (Λ = 2 keV)	fiducial events	ours: median [16–84%]	paper (one event)	our events below the paper's: k / n = fraction ± σ	match
colored 1800 GeV	22	0.19 [0.10, 0.38] ×10^{-3} rad	0.09 ×10^{-3} rad	2 / 22 = 0.09 ± 0.06	**at the low end of our spread**
EW 800 GeV	62	0.52 [0.14, 1.70] ×10^{-3} rad	0.36 ×10^{-3} rad	24 / 62 = 0.39 ± 0.06	inside our 16–84% band



What we learn. The plane turns by less than about 10^{-3} rad over 15 cm in both samples, so the hits stay on one flat plane, as in the paper. The paper's single event lies inside our spread (colored: at its low end).

Figure 4: hits of one signal event

Fig. 4 example event	Δs (plane thickness)	Δw (strip width)	match with the paper
Paper (colored)	83 μm	0.074 cm	—
This work, colored, no smearing	38 μm	0.352 cm	Δs yes (both below the 100 μm cut); Δw no (wider)
This work, EW, no smearing	45 μm	0.100 cm	Δs yes (both below the 100 μm cut); Δw close
This work, colored, 1.16 mm	52 μm	0.470 cm	Δs yes (both below the 100 μm cut); Δw no (wider)
This work, EW, 1.16 mm	73 μm	0.079 cm	Δs yes (both below the 100 μm cut); Δw close

Each example is the signal event with the median Δs. All hits lie on a thin strip, as in the paper.

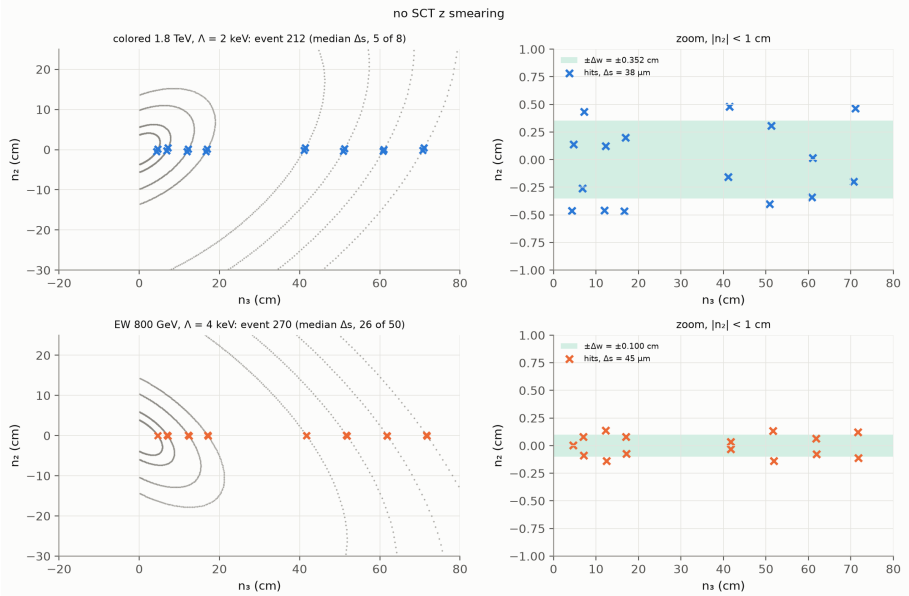
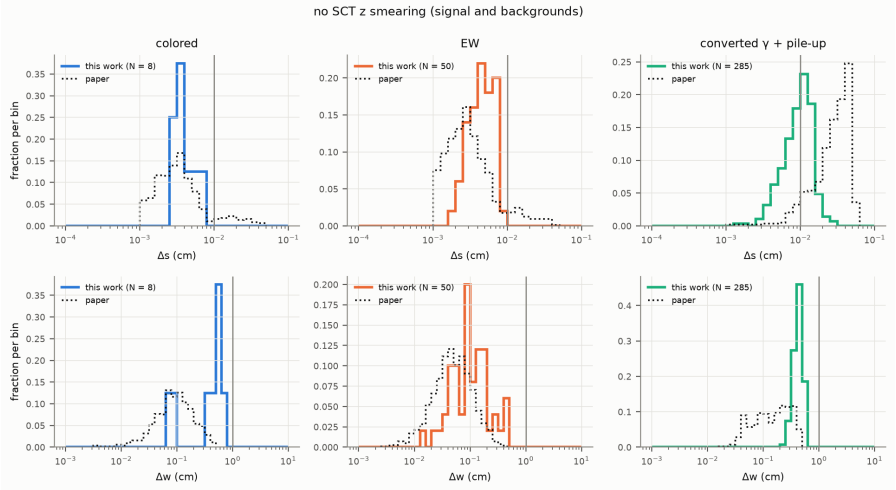


Figure 5: Δs and Δw distributions

Fig. 5	smearing	Δs median: ours / paper	Δw median: ours / paper	fraction with Δs < 0.01 cm: ours / paper	decision
colored signal	none	38 / 32 μm	0.482 / 0.092 cm	8/8 = 1.00 (95% LL 0.69) / 0.90	consistent |Δ| 0.101 ≤ 2σ₀ 0.213
colored signal	1.16 mm	160 / 32 μm	0.550 / 0.092 cm	1/4 = 0.25 ± 0.22 / 0.90	**inconsistent** |Δ| 0.649 > 2σ₀ 0.302
EW signal	none	44 / 28 μm	0.097 / 0.048 cm	50/50 = 1.00 (95% LL 0.94) / 0.92	**inconsistent** |Δ| 0.084 > 2σ₀ 0.078
EW signal	1.16 mm	110 / 28 μm	0.092 / 0.048 cm	10/24 = 0.42 ± 0.10 / 0.92	**inconsistent** |Δ| 0.500 > 2σ₀ 0.113
converted γ + pile-up	none	97 / 306 μm	0.438 / 0.156 cm	146/285 = 0.51 ± 0.03 / 0.07	**inconsistent** |Δ| 0.443 > 2σ₀ 0.030
converted γ + pile-up	1.16 mm	222 / 306 μm	0.350 / 0.156 cm	0/48 = 0.00 (95% UL 0.06) / 0.07	consistent |Δ| 0.069 ≤ 2σ₀ 0.073
pile-up	1.16 mm	317 / 571 μm	0.778 / 0.525 cm	8/109 = 0.07 ± 0.02 / 0.00	**inconsistent** |Δ| 7.2×10^{-2} > 2σ₀ 7.6×10^{-3}



What we learn. Without smearing the signal Δs (≈ 40 μm) is close to the paper's (≈ 30 μm) and all signal passes the Δs cut (paper ≈ 90%), but 0.51 of the converted-photon planes also pass (paper 0.07). With 1.16 mm no converted photon passes, as in the paper, but most signal fails too. No single SCT z setting reproduces both the paper's signal and its background. Δw is about 5× (colored) and 2× (EW) the paper's; it follows the width expected from our own kinematics (arXiv:2002.07503 Eq. III.1: Δw ≈ belt width/√12).

6.4 Why the paper uses Δs < 0.01 cm and Δw < 1 cm (paper reread)

- **Δw < 1 cm: reason given.** Sec. IV: “We deliberately do not impose a tight cut on Δw, as the efficiency for such a cut is strongly signal dependent. The rather loose cut of Δw < 1.0 cm is intended to retain decent efficiencies for quirks with larger oscillation amplitude (low Λ).” The same 1 cm is used in the seeding and in the event selection (Sec. III C), and the paper attributes the low-Λ drop of ε_{reco} “largely” to “a stringent Δw requirement”.
- **Δs < 0.01 cm: no derivation given.** It appears only as “A signal selection cut Δs < 0.01 cm is indicated on the left figure” (Fig. 5), i.e. it is read off Fig. 5, between the signal (≈ 30 μm) and the backgrounds (hundreds of μm). The scale is the expected one: the hits lie on a plane to well within the tracker resolution, which the paper quotes as “roughly ~ 100 μm” (Sec. I). Looser windows are used earlier: seeding Δs < 0.05 cm and $|x \cdot n_1| < 0.05$ cm, growth factor 3 (Sec. III C).
- **General statement** (Sec. III C): “A tighter selection will generally boost computational efficiency at a cost of reduced signal efficiency, which is what has been chosen in this work.”
- **Our numbers:** signal Δs median 38–44 μm without SCT z smearing and 110–160 μm with 1.16 mm; backgrounds 100–600 μm. The value of the cut is therefore not the issue; the SCT z resolution decides whether the signal passes it.

7. Backgrounds

7.1 How many background events pass and fail

Background sample	SCT z	events	plane found (Δw < 1 cm)	+ hit multiplicity	pass (signal region)	fail	paper
converted γ + pile-up (μ = 50)	1.16 mm	9139	1990	48	**0**	9139	“a few” events at 300 fb^{-1}
converted γ + pile-up (μ = 50)	none	9139	4105	285	**146**	8993	“a few” events at 300 fb^{-1}
pile-up only (μ = 50)	1.16 mm	5000	31 (≥ 4 of 8 layers); 1 (all 8)	—	**0**	5000	none in the signal region

Pile-up planes (5000 events, 1.16 mm)	ours: k / n	rate	paper (approx.)	decision
selected plane, ≥ 1 hit in ≥ 4 of 8 layers	31 / 5000	6.2×10^{-3}	$\sim1\times10^{-3}$	**inconsistent** $\lvert\Delta\rvert$ $5.2\times10^{-3} > 2\sigma_0$ 8.9×10^{-4}
selected plane, ≥ 1 hit in all 8 layers	1 / 5000	2.0×10^{-4}	$\sim6\times10^{-5}$	consistent $\lvert\Delta\rvert$ $1.4\times10^{-4} \le 2\sigma_0$ 2.2×10^{-4}

Pile-up only was not simulated without SCT z smearing. Green: no background in the signal region, as in the paper; red: background in the signal region.

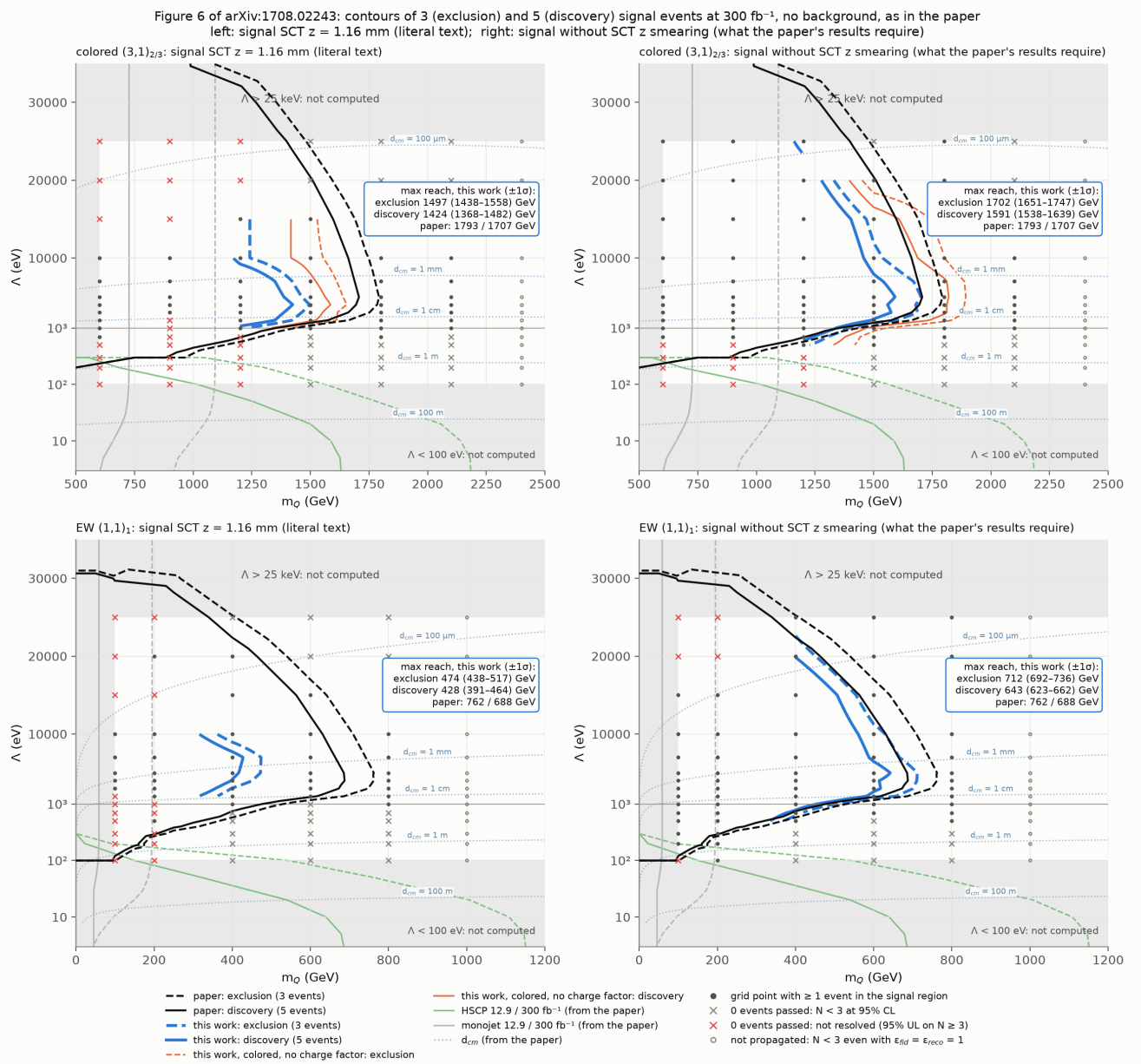
7.2 Converted photons: why the upper limit is the same (Q1)

Converted γ + pile-up (9139 events)	in signal region	per conversion	expected events at 300 fb^{-1}: our σ (0.211 pb) / paper σ (~1 pb)
SCT z 1.16 mm	0 / 9139	< 3.3×10^{-4} (95% UL)	< 0.21 / < 0.98
no SCT z smearing	146 / 9139	0.0160 ± 0.0013	10 / 48

Answer. The limit is a property of our 9139 simulated conversions: 0 pass, so the 95% upper limit per conversion is $1 - 0.05^{1/9139} = 3.3\times10^{-4}$, whatever the cross section. The cross section only sets how many conversions are expected (300 fb^{-1} × σ × 1% = 633 or 3000), which is why the last column differs. Without SCT z smearing on the background, 146 conversions pass and the background is no longer zero.

8. Figure 6: discovery and exclusion contours

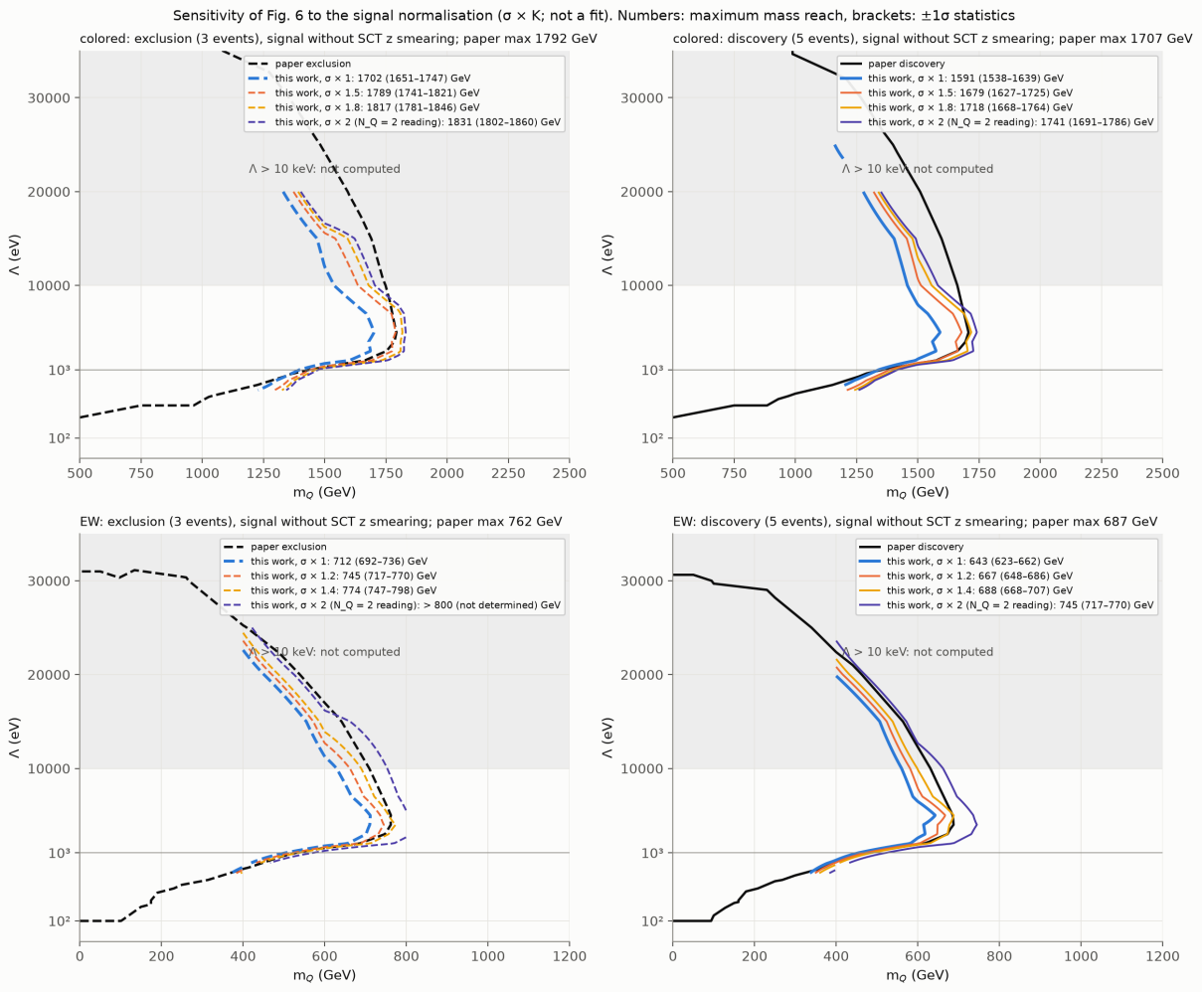
- **Paper:** contours of 3 (exclusion) and 5 (discovery) signal events at 300 fb^{-1}, “assuming negligible irreducible backgrounds”; $N = L\ \sigma\ \varepsilon_{trig}\ \varepsilon_{fid}\ \varepsilon_{reco}$.
- **This work:** the 13 generated samples of figure6_full_v1 (MG5 3.7.3 + Pythia8, MLM-matched, 2000 events per mass: colored 600–2400, EW 100–1000 GeV) through the Table I pipeline at Λ = 0.1–25 keV, SCT z none and 1.16 mm. Colored: matched σ, ε_{trig} by count, both-charged factor f_{ch} included; EW: ε_{trig} = LO ratio $\sigma(Q\bar{Q} + j,\ p_T > 200)/\sigma(Q\bar{Q})$ (0.096 at 800 GeV). LO σ, no K-factor, the “N_Q = 2” of the paper's panel titles not applied.
- **Not computed:** Λ > 25 keV (the integrator's 20-million-step ceiling removes 5–56% of the fiducial events at 30–35 keV), Λ < 100 eV, colored < 600 GeV, EW < 100 GeV; low-Λ edges at light masses are not resolved (0 events passed, red ×).



Normalisation: σ = MadGraph5 3.7.3 LO (0 + 1 jet, MLM-matched with Pythia8) for one quirk species, no K-factor; the paper gives no σ. The "N_Q = 2" of the paper's panel titles is NOT applied (no factor 2 on σ). A factor of 1.5–1.7 (colored) and 1.3–1.4 (EW) on σ brings our maxima close to the paper's (sensitivity, not a fit; F6_kfactor).
EW trigger: ε_trig = σ(pp → QQ̄ + j, p_T(j) > 200 GeV) / σ(pp → QQ̄) at LO (0.096 at 800 GeV, = paper 0.10), NOT the matched + shower count (0.178 at 800 GeV), which would move the EW maximum outward by ~60–80 GeV. EW ε_fid and ε_reco: from the matched + shower sample of each mass; the LO sample at 800 GeV (3000 events, report Section 2.3) gives ε_fid 0.30 vs 0.31 here and k/n within ±13%, which moves the EW maximum by +5 GeV. Colored: ε_trig = matched count; the both-charged factor (~0.33) is included (thin orange: without it).
Red ×: at low Λ for colored 600–1200 GeV and EW 100–200 GeV our samples (40–70 events per mass) are too small; the paper's lower-left edge passes there and is not tested here (larger samples needed). Λ = 15–25 keV: computed (the integrator's 20-million-step ceiling removes ≤ 13% of the events fiducial at 10 keV there); Λ > 25 keV: not computed, because that ceiling removes 5–56% of them at 30–35 keV (needs a larger ceiling and more memory). The maxima are at 3–5 keV and are unaffected.
Annihilation is not simulated (as in the paper). Worst case (α = 0.05, annihilation chance at every second meeting) it lowers the signal near the maxima by ~17–24% (EW 4–5 keV) and ~17% (colored 5 keV), moving the maxima inward by ~30 GeV (colored) and ~25–40 GeV (EW).

Maximum mass reach (GeV), ±1σ	colored exclusion	colored discovery	EW exclusion	EW discovery
paper, Fig. 6	1793	1707	762	688
paper, abstract	–	1600	–	650
this work, no SCT z smearing	1702 (1651–1747)	1591 (1538–1639)	712 (692–736)	643 (623–662)
colored without f_{ch}, no SCT z smearing	1893 (1859–1929)	1820 (1784–1852)		
this work, SCT z 1.16 mm	1497 (1438–1558)	1424 (1368–1482)	474 (438–517)	428 (391–464)
colored without f_{ch}, SCT z 1.16 mm	1655 (1592–1724)	1586 (1527–1651)		

Normalisation sensitivity (no SCT z smearing; not a fit)	colored excl. / disc.	EW excl. / disc.
σ × 1 (our LO)	1702 / 1591	712 / 643
σ × 1.2	–	745 / 667
σ × 1.4	–	774 / 688
σ × 1.5	1789 / 1679	–
σ × 1.8	1817 / 1718	–
σ × 2 (“N_Q = 2” read as a factor 2)	1831 / 1741	> 800 (not determined) / 745
EW ε_{trig} from the matched count (0.18) instead of LO	–	792 / 701
EW ε_{fid}, ε_{reco} from the LO sample at 800 GeV (Section 2.3)	–	717 / 648
annihilation, worst case	1672 / 1560	685 / 605



Result. With the signal SCT z setting that reproduces Table I (no smearing), our contours have the paper's shape: the low-Λ edge near 1 keV, the maximum reach at Λ = 3–5 keV, and the upper part up to 20–25 keV. Our maximum reach (LO σ) matches the paper's abstract (discovery 1600 / 650 GeV) and is 5–7% below its figure. A factor 1.5–1.7 (colored) or 1.3–1.4 (EW) on the signal moves our maxima onto the paper's; the paper gives no σ, so the origin cannot be identified. With the literal SCT z = 1.16 mm the reach is much lower (colored 1424, EW 428 GeV discovery).

9. Figure 7: dE/dx and βγ of the pixel hits (Appendix A)

- **Paper:** “For the dE/dx we use the most probable value as a function of βγ [PDG] … we omitted hits with βγ < 0.1 in the left-hand panel”, benchmarks EW 800 GeV and colored 1800 GeV at Λ = 2 keV. Silicon thickness and the event/hit selection are not stated.
- **This work:** βγ = |p|/m at each pixel-layer crossing (production integrator, Fig. 6 samples at these masses); dE/dx = PDG Eq. 34.12 (Landau–Vavilov–Bichsel most probable value) with the silicon density effect, IBL 200 µm and 250 µm for the other pixel layers (our choice); |z| = 1. Check: 1.11 MeV cm²/g for a minimum-ionising particle in 250 µm (PDG ≈ 1.1).

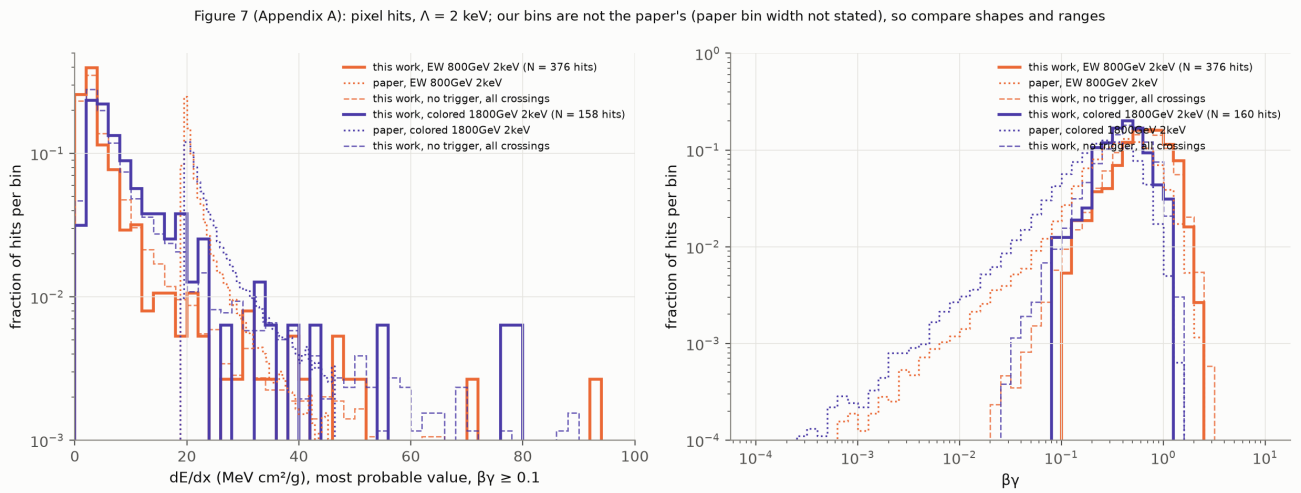


Figure 7 (Appendix A): pixel hits, Λ = 2 keV; our bins are not the paper's (paper bin width not stated), so compare shapes and ranges

Fig. 7 (pixel hits, Λ = 2 keV)	hits	βγ median	hits with βγ < 0.1	dE/dx median (MeV cm²/g)	dE/dx range of the histogram
paper, EW 800 GeV	–	0.35	0.07–0.10	5.6 (PDG MPV of its own βγ histogram)	19–47
this work, EW 800 GeV: triggered, fiducial, first crossing, t < 25 ns	376	0.66	0.000	3.1	1–93
all matched events (no trigger), all crossings, t < 25 ns	8558	0.62	0.012		
all matched events, all crossings, no timing cut	9428	0.61	0.012		
paper, colored 1800 GeV	–	0.22	0.16–0.22	12.6 (PDG MPV of its own βγ histogram)	19–47
this work, colored 1800 GeV: triggered, fiducial, first crossing, t < 25 ns	160	0.42	0.013	6.2	2–79
all matched events (no trigger), all crossings, t < 25 ns	2626	0.43	0.022		
all matched events, all crossings, no timing cut	3059	0.44	0.023		

What we learn. Both distributions peak at βγ ≈ 0.1–1, but our hits are faster: median βγ 0.66 / 0.42 (EW / colored) against the paper's 0.35 / 0.22. Three differences: (1) the paper has a tail of very slow hits (βγ down to 10^{-4}; 7%–10% (EW) and 16%–22% (colored) of its hits below βγ = 0.1, the range given by the bin that contains 0.1); we find at most a few % in any selection (with or without trigger, timing cut or first-crossing requirement). (2) Because our hits are faster, our dE/dx is lower. (3) The paper's dE/dx histogram lies at 19–47 MeV cm²/g for both samples, while the PDG most probable value of its own βγ histogram spans 1–47 with a median of about 6 (EW) and 13 (colored) MeV cm²/g; ours (median 3 and 6) lies in that range and is lower because our hits are faster. The paper does not give enough detail (formula, thickness, units, selection) to trace either point.

10. Open points

- SCT z: the authors' stated resolution gives efficiencies far below Table I; only no smearing reproduces it.
- Signal Δw 2–5× the paper's.
- Fig. 6: our maxima are 5–7% below the paper's figure (they match its abstract); a factor 1.3–1.7 on σ would close the gap; the paper gives no σ.
- Fig. 6: Λ > 25 keV not computed (integrator step ceiling); low-Λ edges at light masses not resolved (larger samples needed).
- Fig. 7: our pixel hits are faster than the paper's (median βγ about 2× higher); the paper's low-βγ tail is not reproduced; its dE/dx range (19–47 MeV cm²/g) is not consistent with its own βγ histogram.
- Paper's Fig. 2: unequal half-oscillations; L definition, charges and momenta of its event not given.

- EW ε_{reco} (no SCT z smearing) above the paper at Λ = 1, 2, 3, 4, 5 keV in the leading-order sample, and similar in the other two EW samples (Section 2.3); the paper gives no information to trace this.

References

Paper: S. Knapen, H. K. Lou, M. Papucci, J. Setford, Phys. Rev. D 96, 115015 (2017), arXiv:1708.02243. [27] ATLAS, ATL-PHYS-PUB-2009-002.
J. Li, T. Li, J. Pei, W. Zhang, “The quirk trajectory”, arXiv:2002.07503 (Eqs. II.11–II.14, A.4–A.5; Sec. VI, App. D).
J. Li, T. Li, J. Pei, W. Zhang, arXiv:1911.02223 (EW trigger 0.1 as an LO ratio, App. A). J. Li et al., arXiv:2108.06748. J. L. Feng et al., arXiv:2404.13814.
Particle Data Group, C. Patrignani et al. (2016), Passage of particles through matter, Eq. 34.12 (most probable energy loss) and Table 34.4 (silicon).
HSCP and monojet limits in Fig. 6: M. Farina, M. Low, private communication to the paper's authors (related: arXiv:1703.00912); taken here from the paper's figure.
Data and scripts: R_data_2026_10_15.py (all numbers), R_report_2026-10-18.py, F38–F42, F6, F7 figures; all plotted numbers as CSV in ROOT_CSV_2026-10-18/.